Yoni Nachmany

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Engineer and leader with research experience and capacity for building technical systems supporting multimodal foundation models, multidimensional data pipelines, and deep learning applications in CV/NLP. Conference speaker.

Experience

Zeus AI — Remote
Head of Data Engineering — May 2024 – May 2025

Zeus AI spun out of NASA to develop ML models for data assimilation and forecasting with Earth observations. As first hire, I built large-scale infrastructure and efficient workflows for generating training data and serving models.

- Operationalized PyTorch foundation model for weather prediction 50x faster than numerical models based on Apple's Massively Multimodal Masked Modeling applied to 4D sensor data, producing a 2249 x 1125 x 37 grid.
- Architected live ingestion/inference pipelines with AWS S3/SNS/Lambda and H100s on Modal/Lambda Labs.
- Scraped, cleaned, and tokenized 50 TBs of video-like satellite imagery with 100+ variables using MPI on HPC.
- Deployed FastAPI demo of hourly changes in solar potential for energy users, visualizing arrays in the browser.

The Linux Foundation — Contract, Remote
Senior Data Scientist — March 2024 – May 2024

The Overture Maps Foundation is an open data collaboration under Linux with dozens of member companies. In the Data Pipeline Task Force, I contributed to the Global Entity Reference System for better data conflation/joining.

- Prototyped Parquet changelogs with PySpark/Databricks for geo data used by Microsoft, Meta, and Amazon.

Gro Intelligence — New York City
Data Scientist, Project Lead — April 2023 – March 2024

Gro Intelligence built an agricultural data platform with predictive models mapping the impact of climate change on commodities. As Data Platform Project Lead, I managed the development of internal data computation/cataloging.

- Orchestrated 33 Airflow DAGs to compute 100+ years of past/projected indices in a versioned array data lake.
- Implemented and oversaw development of internal Python libraries to standardize metadata and input/output.

The New York Times — Contract, New York City
R&D Engineer — December 2021 – September 2022

The NYTimes R&D Team explores emerging technologies and develops new technical capabilities and story formats. I contributed to 3D mapping research and owned the CV/NLP work to create a more accurate digital news archive.

- Scaled transcription of scanned newspapers with multimodal APIs and evaluation framework for 10M articles.
- Implemented 3D satellite stereo reconstruction with photogrammetry libraries to support newsroom coverage.

Mapbox — Washington, D.C.
Software Engineer — September 2019 – November 2021

Mapbox's Imagery team maintains and improves the basemap by sourcing, updating, and enhancing provider data. I extended the capabilities of the data processing pipeline to handle new sources and derive value-added products.

- Updated 135M km^2 of satellite imagery for 3D map launch and saved $1M+ by leveraging open aerial imagery.
- Improved the Mapbox Satellite UX for all users loading data across zoom level groups with novel data fusion.
- Implemented and evaluated deep learning-based object detection/super-resolution (EDSR) for map customers.

Education

University of Pennsylvania — Philadelphia
MSE, Data Science — May 2019
BSE, Networked and Social Systems Engineering — May 2018

Hunter College High School — New York City

Papers

Global Atmospheric Data Assimilation with Multi-Modal Masked Autoencoders	Preprint 2024
Learning Word Ratings for Empathy and Distress from Document-Level User Responses	LREC 2020
Design Considerations for High Impact, Automated Echocardiogram Analysis	ICML 2020
Mitigating Geographic Bias of Image Classifiers with Multilingual Image Data	D4GX 2019
Detecting Roads from Satellite Imagery in the Developing World	CVPR 2019
Generating a Training Dataset for Land Cover Classification to Advance Global Development	NeurIPS 2018
Decoded: Crowdsourcing Summaries of Internet Privacy Policies	HCOMP 2017

Talks

Generating Elevation Data from Stereo Pairs of Satellite Imagery	GeoPython 2023
Depth Perceptions - Perspectives on 3D & Digital Twins	SatSummit 2022
Mapping Historical "Street View" Images of New York City: Visualizing Geotagged Archival Photos	FOSS4G 2022
Under-Mapped Spaces: New Methods and Tools for Critical Storytelling	Stanford 2022
Deep Transfer Learning for Land Cover Classification on Open Multispectral Satellite Imagery	FOSS4G 2019

Internships

Data Science for Social Good Fellow at Imperial College and The Alan Turing Institute	Summer 2019
Machine Learning Intern at Radiant Earth Foundation	Summer 2018
Open Source Software Fellow at Azavea	Summer 2017
Engineering Intern at Meetup	Summer 2016
Machine Learning Research Intern and hackNY Fellow at Clarifai	Summer 2015

Teaching

TA for NETS 213: Crowdsourcing & Human Computation with Prof. Chris Callison-Burch	Spring 2019
TA for ESE 305: Foundations of Data Science with Prof. Victor Preciado	Fall 2017